

TM Forum Component

Product Inventory

TMFC005

Maturity Level: General Availability (GA)	Team Approved Date: 21-May-2026
Release Status: Production	Approval Status: TM Forum Approved
Version 1.1.4	IPR Mode: RAND

Notice

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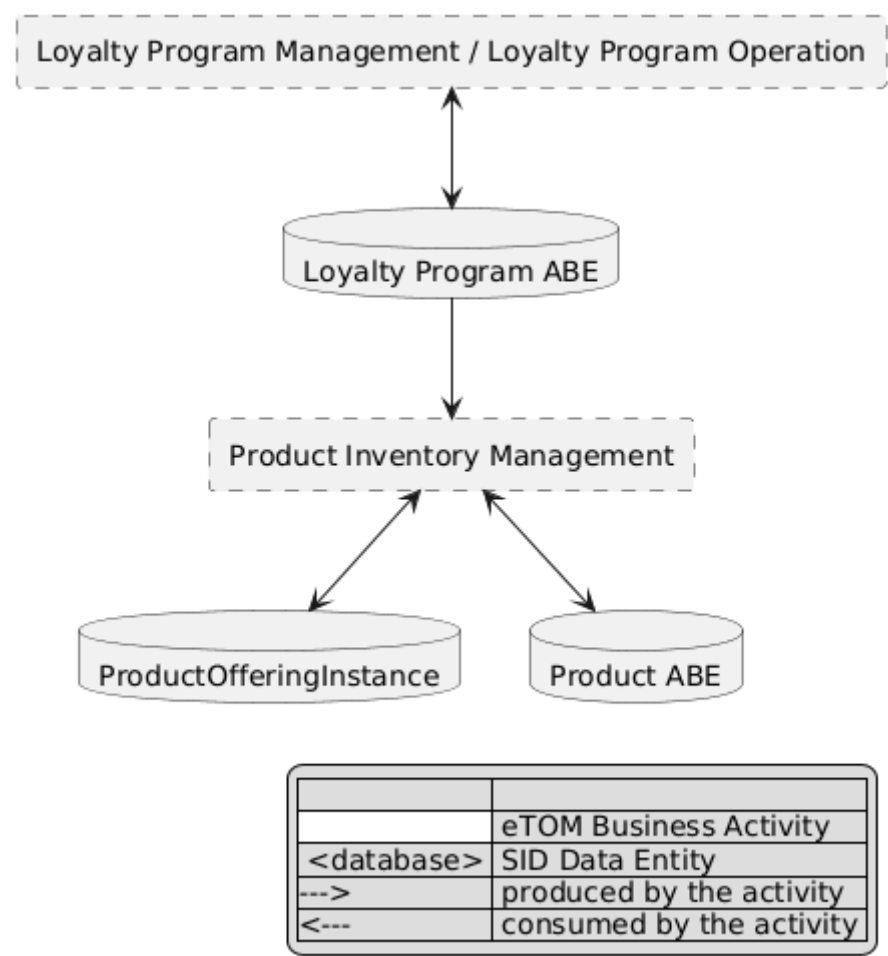
TMFC005 – Product Inventory

1. Overview

Component Name	ID	Description	ODA Function Block
Product Inventory	TMFC005	The Product Inventory component is responsible for storage and exposure of products that are assigned to and used by Parties. This component has functionality that enables creation of inventory items, inventory organization, inventory search or filter, inventory monitoring and tracking, inventory control and inventory auditing. The minimum check to be performed upon inventory item creation or update is for global consistency with related Product Catalog information.	Core Commerce

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.2.11	L2	Product Inventory Management	Product Inventory Management is responsible to establish, manage and administer the enterprise's product inventory, as embodied in the Product Inventory repository, and monitor and report on the usage and access to the product inventory.

1.1.19	L2	Loyalty Program Management	Define all aspects of a loyalty program, such as requirements and objectives of a loyalty program, determine the benefits to participants. Develop a program, prototype it, test it, rollout/launch it, amend and evaluate it, and terminate it when it is no longer viable for an enterprise.
1.1.19.2	L3	Loyalty Program Operation	Manage all operational aspects of running a loyalty program. Enable parties to become a members of a program, earn currency and rewards, and redeem currency. Manage a loyalty program account, leave a program, and provide operational reports.
1.1.19.2.5	L4	Manage Loyalty Program Account	Update a loyalty program account and make changes to loyalty program participant information. Expire, reinstate, transfer in/out, adjust, a loyalty participant's account currency. Prepare and send a loyalty program communication to a participant or for internal use by an enterprise.
1.1.19.2.7	L4	Provide Loyalty Program Operation Report	Generate a loyalty program operation report, such as various loyalty program status reports, trend analysis, and reports that identify suspected abuse of a loyalty program.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
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Product and Offering Instance	Product and ProductOfferingInstance represent, in the product inventory, respectively an instance of a ProductSpecification and ProductOfferingSpecification subscribed or not yet subscribed to by a Customer from the CSP or by the CSP from a BusinessPartner. The Product carries the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Product and Offering Instance ABE also tracks the links to services and/or resources through which the Product is realized as well as the applied tariff and commitment duration related to the ProductOfferingInstance.	Product	Represents an instance in the product inventory of a ProductSpecification subscribed or not yet subscribed by a Customer to the CSP or by the CSP to a BusinessPartner, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. It may come from a CustomerProductOrderItem or a BusinessPartnerOrderItem. At one time, it might be contained in a unique ProductOfferingInstance. The Product ABE also tracks the links to services and/or resources through which the product is realized. The Product ABE also contains detailed information concerning Network Products and Usage Volume Products as well as other examples of Products.
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Product and Offering Instance	Product and ProductOfferingInstance represent, in the product inventory, respectively an instance of a ProductSpecification and ProductOfferingSpecification subscribed or not yet subscribed to by a Customer from the CSP or by the CSP from a BusinessPartner. The Product carries the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Product and Offering Instance ABE also tracks the links to services and/or resources through which the Product is realized as well as the applied tariff and commitment duration related to the ProductOfferingInstance.		(no description available)
Loyalty	A loyalty program is one of the tools used by the loyalty process to retain customers. The Loyalty ABE contains all entities useful to specify and instantiate loyalty programs. A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.	Loyalty Program	The Loyalty Program ABE contains all entities needed to instantiate a loyalty program for a specific customer such as LoyaltyProgram and LoyaltyAccount.

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
180	Assigned Products Maintenance	Product Management	Product Repository Management	Assigned Products Maintenance permits defining and update: / - product characteristics / - links with the related service or resource (handsets, SIM cards, ...) needed to deliver the product, ...
197	Customer Product Storage	Product Management	Product Repository Management	Product Storage provides the functionality necessary to store and make available the Products. / This function allows: / • to instantiate or update offers and products ordered by the customer, whatever their type (network product, bundle, device, ...) or their marketing mode (rented, sold, ...), with their configuration, their tariffs and discounts, and their status (initialized with a creation order) / • to update Products status / • to search and read Offer and Product installed base (subscribed offers, configuration of installed products, installed tariffs and discount, statuses, ...).

198	Customer Loyalty Score Balance Management	Product Management	Loyalty Account Management	Customer Loyalty Score Balance Management function calculates the score according to accumulation/decrease rules. When a customer subscribes to the loyalty program with more than one SIM or other 'traffic objects', the Score Management accumulates the points into a single balance. The loyalty score could decrease for one of the following events: prize purchase, points expiry or points deletion by Call Centre. The functionality may also include the visualization of Score details (date, description event type, points, final score) via different contact Channels (e.g. via Web, IVR, Call Centre).
237	Customer Loyalty Communication	Product Management	Loyalty Account Management	Customer Loyalty Communication function sends information related to Loyalty Programs (Point Balance, Prize Request status, renewed Loyalty Code) to external components in push and pull modes.

361	Contract Implementation	Product Management	Product Agreement Management	Product Agreement Implementation function provides functionality pertaining to the implementation of the Product Agreement (a.k.a. contract) across fulfillment, assurance, and billing according to Product Agreement Specification. / A Product Agreement represents the approval by the Customer and the Vendor of all term or conditions of a ProductOffering.
362	Contract Searching	<i>(no description available)</i>	<i>(no description available)</i>	<i>(no description available in the source document or the Functional Framework spreadsheets)</i>

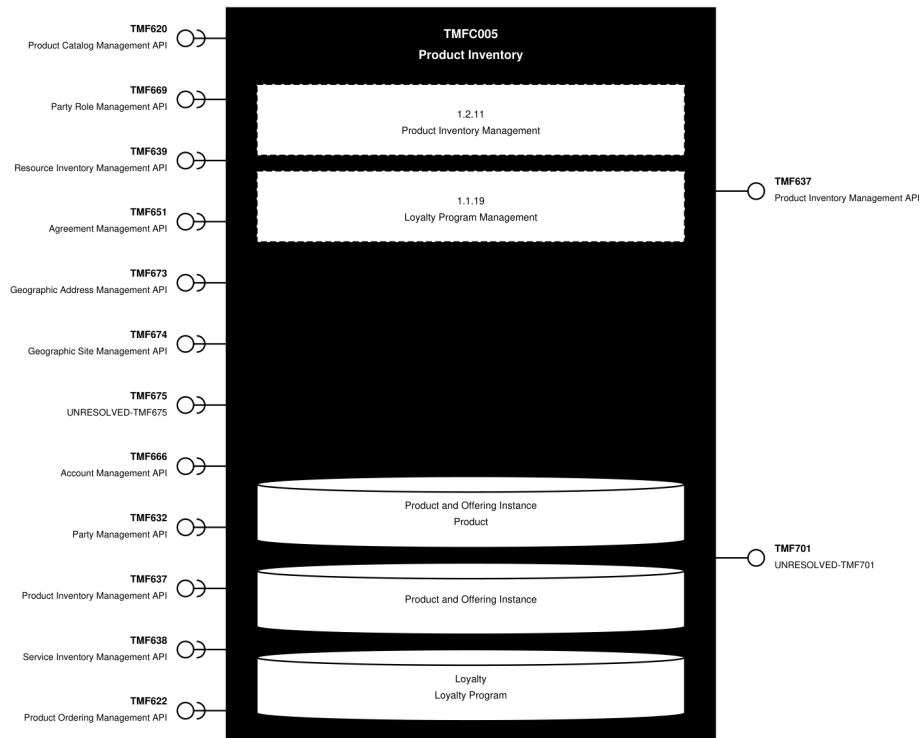
363	Contract Storage	Product Management	Product Agreement Management	Contract Storage provides the central repository for contract storage as well as the associated contract meta-data. This data can be mined for Campaigns and Lead Generation. / Product Agreement Storage provides functionality necessary to store and make available the Product Agreements. / This function allows: / • to instantiate or update Product Agreement approved by the customer with their party involved, their configuration, their approvals and their status, / • to update Product Agreements status, / • to search and read Product Agreements.
1201	Product Configuration Check	Product Management	Product Repository Management	Product Configuration Check Function Checks for each Product submitted if all Product configuration rules have been respected such as prerequisite, incompatibility rules between ProductSpecifications or mandatory characteristics.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events

(generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

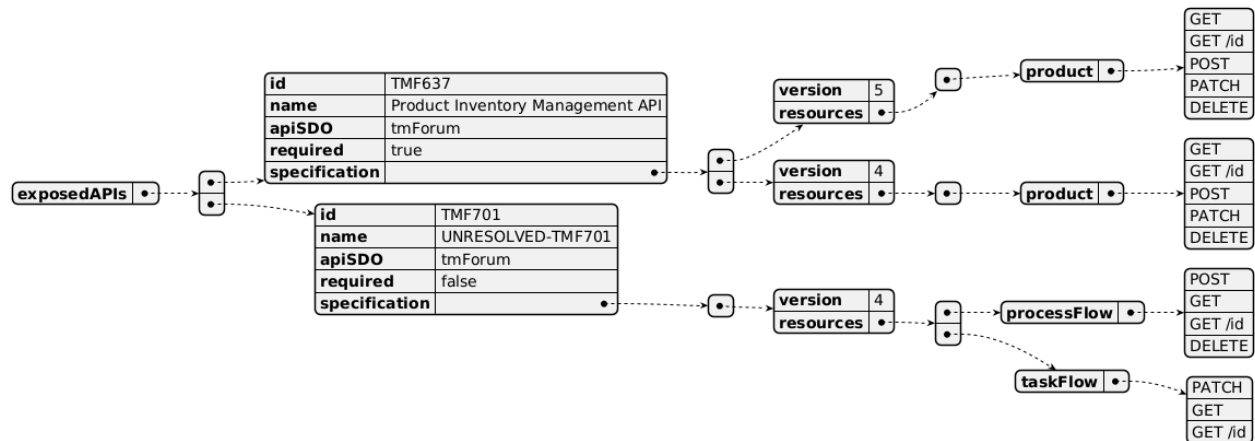


(SVG source: [TMFC005 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the component-specification-documentation skill's `scripts/render_api_context_svg.py`.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF637	Product Inventory Management API	Mandatory	5	product	GET, GET /id, POST, PATCH, DELETE
TMF637	Product Inventory Management API	Mandatory	4	product	GET, GET /id, POST, PATCH, DELETE

TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id

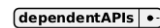


3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF620	Product Catalog Management API	5	Mandatory	productOffering	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productSpecification	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productSpecification	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productOffering	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productOfferingPrice	GET, GET /id
TMF669	Party Role Management API	5	Optional	partyRoleManagement	GET, GET /id

TMF669	Party Role Management API	4	Optional	PartyRoleManagement	GET, GET /id
TMF639	Resource Inventory Management API	4	Optional	resource	GET, GET /id
TMF651	Agreement Management API	4	Optional	agreement	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicSubAddress	GET, GET /id
TMF674	Geographic Site Management API	4	Optional	geographicSite	GET, GET /id
TMF675	UNRESOLVED-TMF675	4	Optional	geographicLocation	GET, GET /id
TMF666	Account Management API	5	Optional	billingAccount	GET, GET /id
TMF666	Account Management API	4	Optional	billingAccount	GET, GET /id
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF637	Product Inventory Management API	5	Optional	product	GET, GET /id, POST, PATCH, DELETE
TMF637	Product Inventory Management API	4	Optional	product	GET, GET /id, POST, PATCH, DELETE
TMF638	Service Inventory Management API	5	Optional	service	GET, GET /id

TMF638	Service Inventory Management API	4	Optional	service	GET, GET /id
TMF622	Product Ordering Management API	5	Optional	productOrder	GET, GET /id
TMF622	Product Ordering Management API	4	Optional	productOrder	GET, GET /id

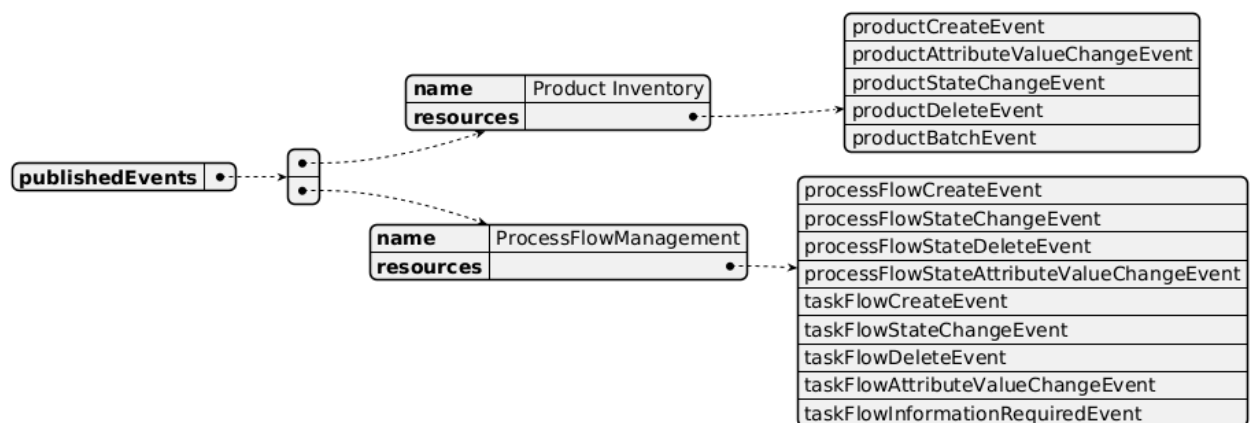


3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

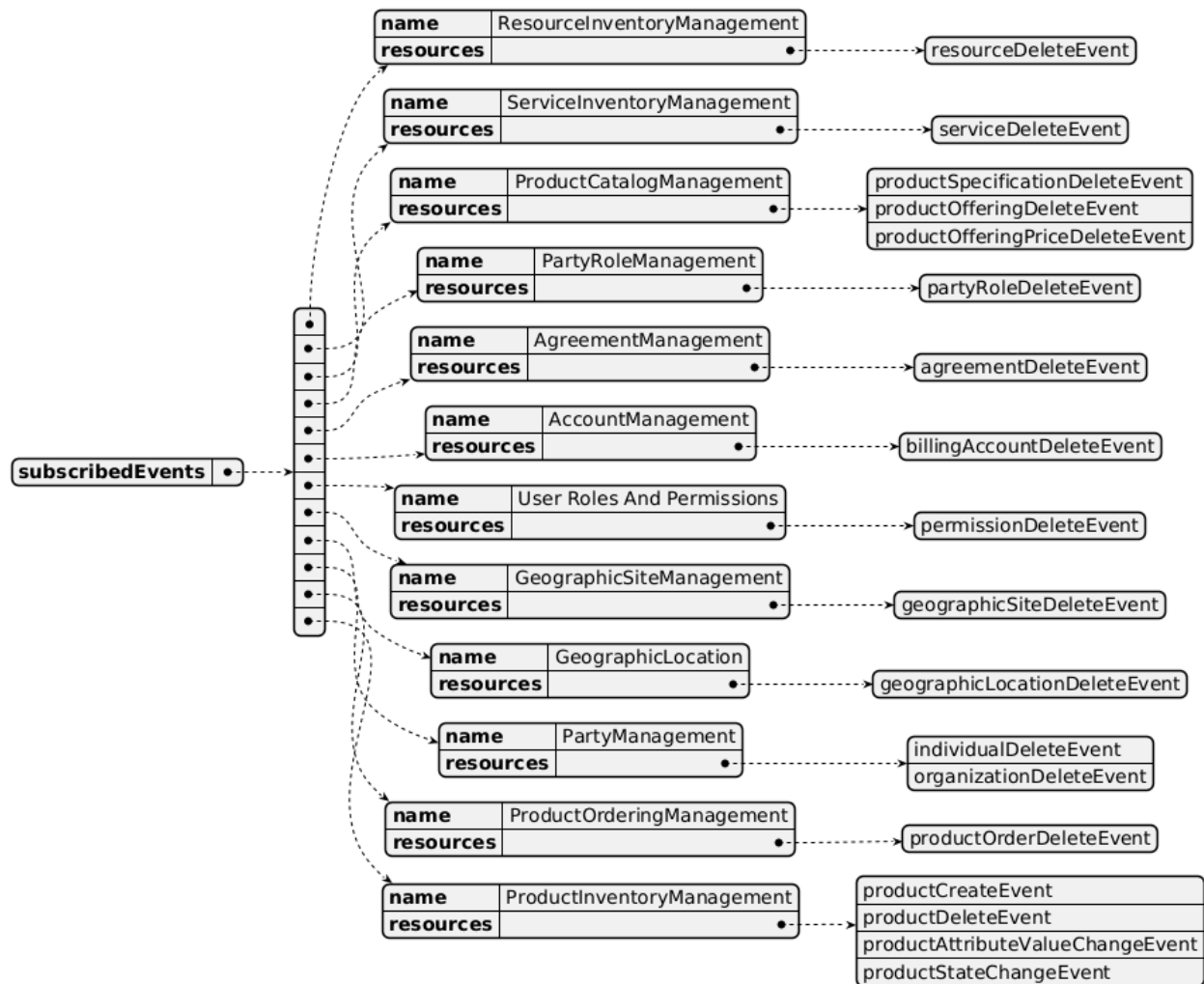
API ID	API Name	Event Resources
TMF637	Product Inventory Management API	productCreateEvent productAttributeValueChangeEvent productStateChangeEvent productDeleteEvent productBatchEvent
TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent



Subscribed Events

API ID	API Name	Event Resources
TMF639	Resource Inventory Management API	resourceDeleteEvent

TMF638	Service Inventory Management API	serviceDeleteEvent
TMF620	Product Catalog Management API	productSpecificationDeleteEvent productOfferingDeleteEvent productOfferingPriceDeleteEvent
TMF669	Party Role Management API	partyRoleDeleteEvent
TMF651	Agreement Management API	agreementDeleteEvent
TMF666	Account Management API	billingAccountDeleteEvent
TMF672	User Role Permission Management API	permissionDeleteEvent
TMF674	Geographic Site Management API	geographicSiteDeleteEvent
TMF675	UNRESOLVED-TMF675	geographicLocationDeleteEvent
TMF632	Party Management API	individualDeleteEvent organizationDeleteEvent
TMF622	Product Ordering Management API	productOrderDeleteEvent
TMF637	Product Inventory Management API	productCreateEvent productDeleteEvent productAttributeValueChangeEvent productStateChangeEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v24.0
SID	v25.0

Functional Framework	v24.0
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5.2. Jira References

eTOM - <https://projects.tmforum.org/jira/browse/ISA-413> L2 - Loyalty Program Management (1.1.19): review positioning of L2 (why in Market & Sales and not in Product) and detailed L3 and L4

SID (*none currently listed*)

Functional Framework - <https://projects.tmforum.org/jira/browse/ISA-415> Add a function to check data integrity at Product level, including mandatory relationships. -

<https://projects.tmforum.org/jira/browse/ISA-414> Clarify functions using "contract" in their name and/or definition as globally at product level (instantiation of Product Offering or Product Specification). - <https://projects.tmforum.org/jira/browse/ISA-416> Harmonize as much as possible functions related to Product, Service and Resource Inventory

API - TMF637 - Product Inventory Management: list missing information and relationship as Loyalty Account - <https://projects.tmforum.org/jira/browse/AP-3844> -

<https://projects.tmforum.org/jira/browse/AP-3139> - TMF637 Product Inventory Management: list missing information related to SID ABE productOfferingInstance -

<https://projects.tmforum.org/jira/browse/AP-7096> - ProductPriceCharge should be managed by this component, including all specialization of ProductPriceCharge (oneTime, recurring, usage...)

- See <https://projects.tmforum.org/jira/browse/ISA-1369>

5.3. Further resources

This component is involved in the following use cases described in IG1228 How to use ODA - Using Open APIs to realize Use Cases.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
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1.0.0	31-May-2022	Kamal Maghsoudlou, Sylvie Demarest	Final edits prior to publication
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.0.2	11-Jun-2024	Sylvie Demarest	Updated Component Template. Aligned to Frameworks 23.5
1.0.3	28-Mar-2025	Gaetano Biancardi	Correct typo in Exposed API diagram
1.0.3	08-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025. Final edits prior to publication
1.0.4	30-Sep-2025	julien rouland	Update to reflect productOfferingInstance ABE as mentioned in ISA-414 - Clarify Product Inventory Mgt sub-domain vocabulary SME REVIEW
1.0.4	19-Nov-2025	Rosie Wilson	Final edits prior to publication
1.0.4	16-Jan-2026	Adrienne Walcott	Updated to reflect TM Forum Approved status
1.1.0	28-April-2026	Hugo Vaughan (TM Forum)	Gen 5 API incorporation
1.1.0	25-Jun-2025	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	31-May-2022	Goutham Babu	Initial release
Production	22-Jul-2022	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	14-Aug-2023	Amaia White	New release 1.0.1

Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.0.2
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	08-Jul-2025	Rosie Wilson	New release 1.0.3
Production	05-Sep-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	19-Nov-2025	Rosie Wilson	New release 1.0.4
Production	16-Jan-2026	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	01-Jun-2026	Sannah Sibaya	New release 1.1.0; final edits prior to publishing

6.2. Acknowledgements

This document was prepared by the members of the TM Forum ODA Components & Canvas team.

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